

Establishing homology between monkey and human brains

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Neuroimaging methods are beginning to provide promising ways of understanding the functional organization of the brain across species.

Recent decades have seen a massive scientific effort to identify the functional organization of the brain in both humans and non-human species. Work in humans and other species provides complementary insights: animal models provide unique opportunities for invasive study of biological mechanisms, but differences in brain organization across species are a major obstacle to the identification of functional homologies. As a result, human and animal work is all too often discussed in parallel, separate literature. In this issue of *Nature Methods*, Mantini and colleagues introduce a method for identifying cross-species homologies to the neuroscientist's arsenal¹.

Their technique, interspecies activity correlation (ISAC), uses functional magnetic

resonance imaging (fMRI) to identify brain regions in which humans and monkeys exposed to the same dynamic stimulus—a 30-minute clip from the movie *The Good, the Bad and the Ugly*—show correlated patterns of activity¹ (Fig. 1). The premise is that homologous regions should have similar patterns of activity across species. For example, a brain region sensitive to a particular configuration of features, including visual motion, hands, faces, object and others, should show a similar time course of activity in both species—even if its anatomical location differs across species and even if the precise features that drive the area's neurons have not yet been specified.

Previously, homology has been established primarily based on anatomical and

cytoarchitectural similarity² or on functional responses to specific stimuli (for example, motion in a particular part of the visual field^{3,4}). However, such work is complicated by the absence of a simple 1:1 mapping across species, the great number of possible homologies and the laborious nature of such mapping experiments. The ISAC method is a sort of 'high-throughput' alternative that allows the simultaneous mapping of potential homologous regions across the brain. It does so without reference to particular classes of stimuli, identifying homologous regions that respond to as-yet undefined classes of neural processes, for example, regions that respond to complex combinations across multiple features (hands, objects, faces and others), interactions between visual features and other 'top-down' cognitive processes. In combination with established homologies, the ISAC method is a potentially valuable and important way of synthesizing findings across species.

As with any new method, there are challenges to be overcome. One important obstacle is that the ISAC method¹ requires that humans and monkeys engage the same psychological processes during stimulus viewing. For highly dynamic stimuli such as movies, this is often not the case: humans view a movie in the context of its narrative and meaning, which can influence the strength and quality of attention, eye movements and fMRI activity throughout the cortex and thalamus, including 'early' visual areas⁵. Consequently, some homologies may be missed and some false homologies may be identified.

Homologies will be missed when the same process is engaged at different times in each species—for example, if humans pay attention to facial cues that monkeys are oblivious to. False homologies could be identified when the same stimulus engages attention to different aspects of complex stimuli—for instance, Clint Eastwood's appearance on screen is associated with both low-level features (for example, faces and gross body movements) as well as multimodal and high-level ones (for example, speech and facial affect). Monkeys might attend primarily to Eastwood's gross movements, whereas humans might attend to his facial expressions and speech, causing the ISAC method

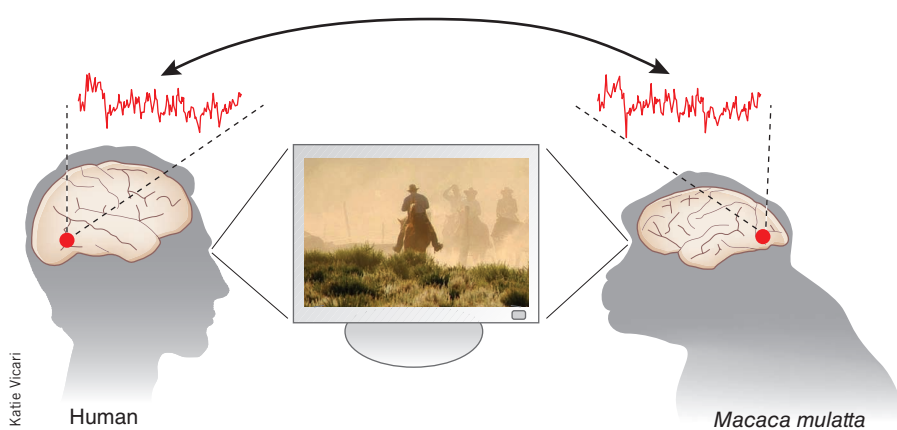


Figure 1 | The ISAC method. Humans and monkeys watch the same film as their brains are scanned with fMRI. The time course of fMRI activity during viewing is extracted from 'seed' regions in human participants and correlated with the fMRI signal in monkeys, and vice versa, with an adjustment made for interspecies differences in vascular hemodynamics. After statistical thresholding, the resulting maps in each species show areas that are potentially homologous with the targeted seed region in the other species.

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to identify a spurious ‘homology’ between regions responsive to motion and body parts in monkeys and faces, prosody and social cognition in humans. This phenomenon provides a rival explanation for some of the putative homologies Mantini *et al.*¹ identify across anatomically divergent regions.

The susceptibility of the ISAC method to these pitfalls will no doubt be investigated in future studies, as will its ability to provide information on homology outside the visual system. Subsequent studies could replicate interspecies activity correlations across different types of stimulus sets, including those that monkeys and humans are more likely to perceive in similar ways. In addition, putative homologies could be corroborated by follow-up fMRI or electrophysiological studies attempting to find the critical features that drive regional activity across species.

Another exciting future direction is the hybridization of the ISAC method with

established techniques for retinotopic and functional mapping so that candidate homologous regions may be jointly constrained by interspecies activity correlations and similarity in functional responses. Ultimately, this type of approach could be used to integrate other types of information as well, including patterns of gene expression and cytoarchitecture. This strategy may be supplemented by methodological developments. Multivariate methods⁶ can be used to both bypass the need for a priori definition of ‘seed’ regions based on anatomy and simultaneously consider multiple sources of information. Pattern-based methods could be used to identify regions that code for similar types of information across species⁴.

In sum, the ISAC approach will lead to new opportunities, new challenges and new conflicts to be resolved, as any new technique does. It comes at a crucial and exciting time in the neurosciences, as new techniques

including fMRI can finally be used to directly map brain function in humans and other species, and this wealth of parallel information must be integrated to bring insights from animal models to bear on the human condition in increasingly precise ways.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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